ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)

ORGANISATION OF ISLAMIC COOPERATION (OIC)

Department of Computer Science and Engineering (CSE)

QUIZ 3 SET D **SUMMER SEMESTER, 2023-2024**

DURATION: 30 MINUTES **FULL MARKS: 20**

CSE 4205: Digital Logic Design

Programmable calculators are not allowed. Do not write anything on the question paper.

Answer **all 3 (three)** questions. Figures in the right margin indicate full marks of questions with corresponding COs and POs in parentheses.

ID:

1. You have to design a $2^3 \times 3$ priority encoder with a Valid output. The encoder has 8 inputs labeled I_0 to I_7, 3 output lines labeled O_2, O_1, O_0, and a Valid signal V. 4.5 (CO5) (PO3)

   The inputs have the following randomly assigned priority (from highest to lowest):

   $$\text{Priority: } I_2 > I_5 > I_1 > I_7 > I_3 > I_0 > I_6 > I_4$$

   To be noted -

   1. If multiple inputs are **HIGH**, the encoder outputs the binary code of the **highest-priority input**.
   2. The **Valid** output V is 1 if any **input** is **HIGH**, and **0** if **all are LOW**. You can arrange the inputs however you like.

   Fill in the truth table based on this priority.

Inputs								**Outputs**			
								O_2	O_1	O_0	V

2. Design a combinational circuit using comparator ICs and logic gates to determine the **smallest among three input numbers**: A, B, and C. The circuit must satisfy the following specifications: 7 + 1 + 2 (CO5) (PO3)

- **Inputs:** Three binary numbers A, B, and C (you are not required to consider the number of bits, assume they are compatible with the comparator IC).
- **Outputs:** Three output signals:
  - **A smallest**: High (1) if $A < B$ and $A < C$
  - **B smallest**: High (1) if $B < A$ and $B < C$
  - **C smallest**: High (1) if $C < A$ and $C < B$
- It is guaranteed that no two inputs are equal. Hence, exactly one of the outputs will be high at a time.
- **Comparator IC:** You may only use 2-input comparator blocks. Each comparator has two n-bit number inputs (X and Y) and three outputs: $X > Y, X < Y$, and $X = Y$, where X and Y are two n-bit numbers.

  You are not required to design the comparator IC, it may be represented as a block in your diagram.
- **Design Requirement:** Using these comparator IC blocks and any other component as you wish, draw the **block diagram** of a combinational circuit that produces the required outputs. You may assume ideal behavior and unlimited availability of logic gates and comparators.

Note: A block diagram means that you do not need to draw the internal structure of the comparator ICs. Instead, use labeled blocks showing inputs and outputs clearly.

Answer the following questions -

a) Design the above-mentioned circuit.

b) Explicitly mention the number of 2-number comparator ICs you have used.

c) Briefly explain the working mechanism (avoid unnecessary details).

d) Propose the most efficient solution in terms of the number of comparators used. **[Bonus 5 Marks]**

3. Given, 3.5 + 2

(CO5) (PO3)

$$F_1(A,B,C) = \sum m(3,4,7)$$
$$F_2(A,B,C) = \sum m(2,3,6)$$

Complete the PLD circuit for F_1 and F_2 by making the necessary connections (using 'X' at junctions) using the following **PLA (Programmable Logic Array)** structure.

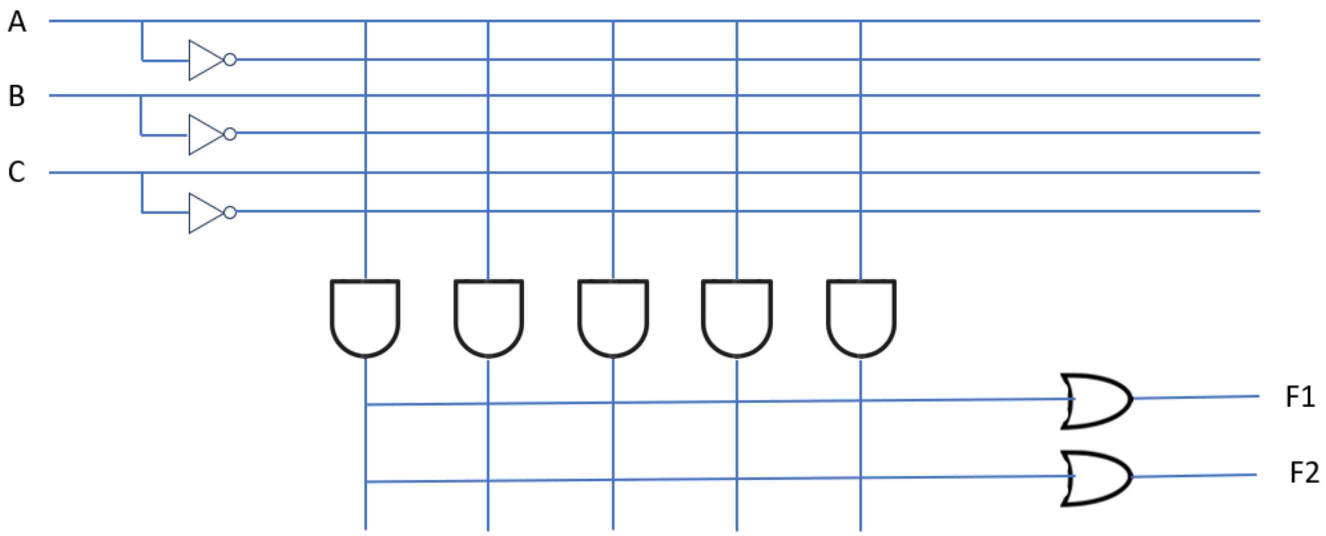

Figure 1: PLA Circuit

And answer the following questions-

a) A PLA circuit structure is defined by $n \times p \times m$. Write down what each symbol represents and provide their values according to the given PLA circuit diagram that you have constructed.

- n: ____, **Value:** __________
- p: ____, **Value:** __________
- m: ____, **Value:** __________

b) Put a tick next to the correct option.
For a **ROM (Read Only Memory)**:

- AND Array: ☐ Fixed ☐ Programmable
- OR Array: ☐ Fixed ☐ Programmable